Analysis of **wildfire** visualization systems for research and training: are they up for the challenge of the current state of wildfires?

CAT Cortes, S Thurow, A Ong… - IEEE transactions on …, 2023 - ieeexplore.ieee.org

… **survey** reviews various studies that use immersive and non-immersive data visualization techniques to depict **wildfire behavior** … can help **wildfire management** offices during a **wildfire** …

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Cell2Fire: A cell-based **forest fire** growth **model** to support strategic landscape **management** planning

C Pais, J Carrasco, DL Martell, A Weintraub… - Frontiers in Forests …, 2021 - frontiersin.org

… **model** in which fire **spread** is based on existing fire **spread** models developed by others (in our case, the Canadian Fire **Behavior prediction** … models, and a brief **review** of the fire growth …

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The use of science in **wildland fire management**: a **review** of barriers and facilitators

ME Hunter, MM Colavito, V Wright - Current Forestry Reports, 2020 - Springer

… fire **behavior** and meet other resource **management** objectives. We consider **wildfire management** as … , or the monitoring or facilitation of **spread** of naturally caused wildfires if they are …

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Atmospheric interactions with **wildland fire** behaviour–I. Basic surface interactions, vertical profiles and synoptic structures

BE Potter - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… rate and wind speed based on the results of their numerical **model** with coupled fire … Part II of this **review** discusses plume structure and **dynamics** in detail, including elaboration on …

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Wildfire spread prediction using attention mechanisms in u2-net

H Xiao, Y Zhu, Y Sun, G Zhang, Z Gong - Forests, 2024 - mdpi.com

… We categorize this **wildfire spread prediction** task as a semantic segmentation problem in image segmentation, where each region is predicted to contain a fire or not based on the …

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Rapid **prediction** of **wildfire spread** using ensemble Kalman filter and polyline simplification

S Yoo, J Song - Environmental Modelling & Software, 2023 - Elsevier

… This paper proposes a new method for rapid **prediction** of **wildfire spread**, which employs computational **wildfire** simulations by FARSITE and assimilates the **simulation** results with …

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Some requirements for simulating **wildland fire behavior** using insight from coupled weather—**wildland fire** models

J Coen - Fire, 2018 - mdpi.com

… **Wildland fire behavior** models have been developed and … environmental conditions on fire **behavior**. Early models were … fire **behavior simulation** results to advise potential users on …

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Remote Sensing for **Wildfire** Mapping: A Comprehensive **Review** of Advances, Platforms, and Algorithms

RE Guiop-Servan, A Cotrina-Sanchez, J Puerta-Culqui… - Fire, 2025 - mdpi.com

… **review**, including terms such as “forecasting” and “modeling”, which, although not strictly limited to **wildfire** … involve fire risk assessment, fire **spread prediction**, or related remote sensing …

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Autonomous unmanned aerial vehicles in **bushfire management**: Challenges and opportunities

S Partheepan, F Sanati, J Hassan - Drones, 2023 - mdpi.com

… This **review** focuses on the use of UAVs in **bushfire management** for fire detection, fire … named Spark for **bushfire spread** predictions. Users can design their fire **propagation** models by …

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Estimation of **wildfire** size and risk changes due to fuels treatments

MA Cochrane, CJ Moran, MC Wimberly… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… modelling system (Finney 2004) to simulate the observed **wildfire** progression and **spread** rates. **Model** simulations were parameterised with LANDFIRE fuels and topographic data …

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